Calibrating Noise to Sensitivity for Differential Privacy (Dwork et al. 2006)

Differential privacy bounds the privacy loss of individual data participants by calibrating additive noise to the global sensitivity of query functions, providing mathematical guarantees against data reconstruction.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Dwork et al. (2006) formalize differential privacy, providing a mathematical guarantee that the presence or absence of any single individual in a database does not significantly alter the probability distribution of any computational output. By calibrating Laplacian noise to the global L_1 sensitivity of the query function ($\Delta f / \varepsilon$), the mechanism ensures ε-differential privacy, mathematically precluding reconstruction and linkage attacks regardless of auxiliary background knowledge.

EVIDENCE AND CAUSAL ASSESSMENT

Mathematical proof establishes that bounded noise injection bounds the privacy loss random variable by e^{ε}, guaranteeing semantic plausible deniability for every data contributor.

EFFECT SIZE AND PERSISTENCE

Eliminates the vulnerability of heuristic anonymization (k-anonymity, data swapping) to external linkage attacks, establishing absolute bounds on information disclosure that persist under arbitrary post-processing.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as n/a (formal mathematical proof). Applying small privacy budgets ($\varepsilon < 1$) requires substantial additive noise that can distort downstream statistical utility, particularly in small subgroup populations.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Differential privacy does not guarantee absolute secrecy of population-level facts discovered from aggregate data; it guarantees that an individual's personal participation cannot be reliably inferred.

WHAT ACTION IS REASONABLY JUSTIFIED

Mandate differential privacy frameworks (such as DP-SGD or randomized response mechanisms) across sensitive population telemetry and machine learning training pipelines.

VERIFIED SOURCES & DOI IDENTIFIERS

- Dwork, C., McSherry, F., Nissim, K., & Smith, A. (2006). Calibrating Noise to Sensitivity in Private Data Analysis. In Theory of Cryptography (pp. 265–284). Springer, Berlin, Heidelberg. https://doi.org/10.1007/11681878_14